⍰ ATOMIC STRUCTURE

Introduction

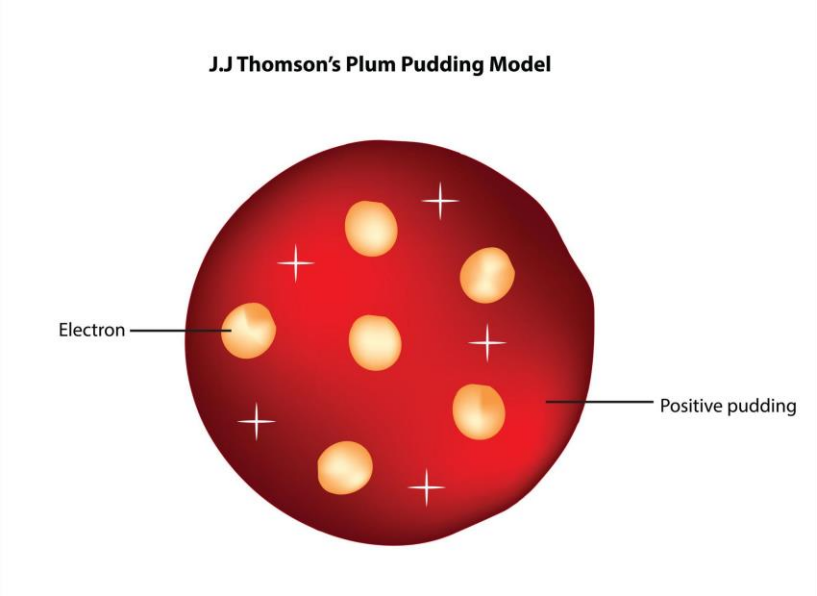
Atomic structure deals with the internal composition of atoms, including electrons, protons, and neutrons. Understanding atomic structure is essential for explaining chemical behavior, bonding, and periodic properties. This chapter forms the foundation of chemistry and is highly important for NEET.

Discovery of Subatomic Particles

The atom was initially considered indivisible, but later experiments revealed the existence of subatomic particles. Electrons were discovered by J.J. Thomson through cathode ray experiments. Protons were identified by Goldstein, and neutrons were discovered by Chadwick. These discoveries helped in understanding the internal structure of atoms.

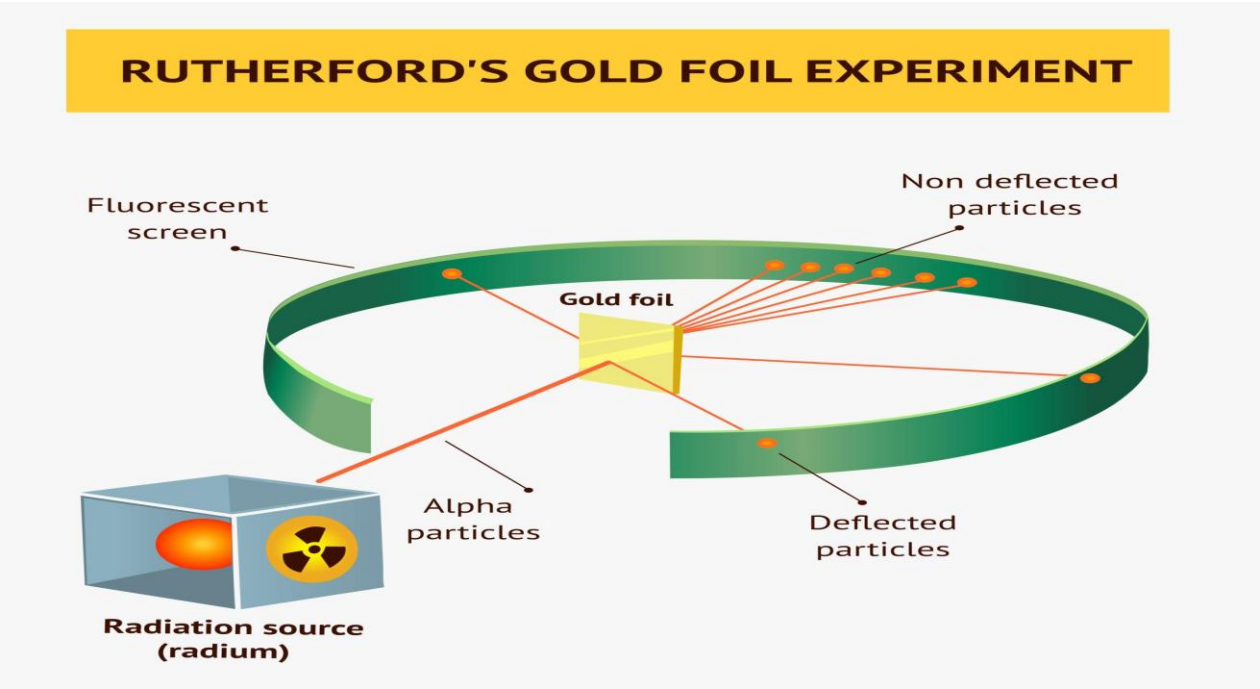
Thomson's Atomic Model

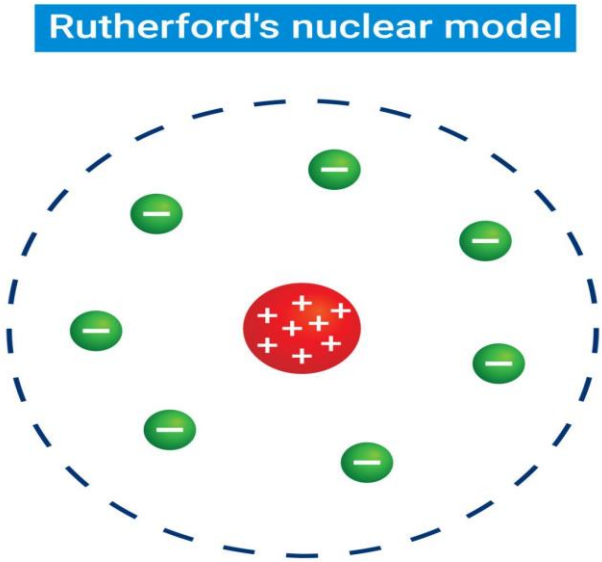
Thomson proposed that an atom is a positively charged sphere with electrons embedded in it, similar to a "plum pudding" model.



Rutherford's Atomic Model

Rutherford's gold foil experiment showed that most of the atom is empty space with a small, dense, positively charged nucleus at the center.

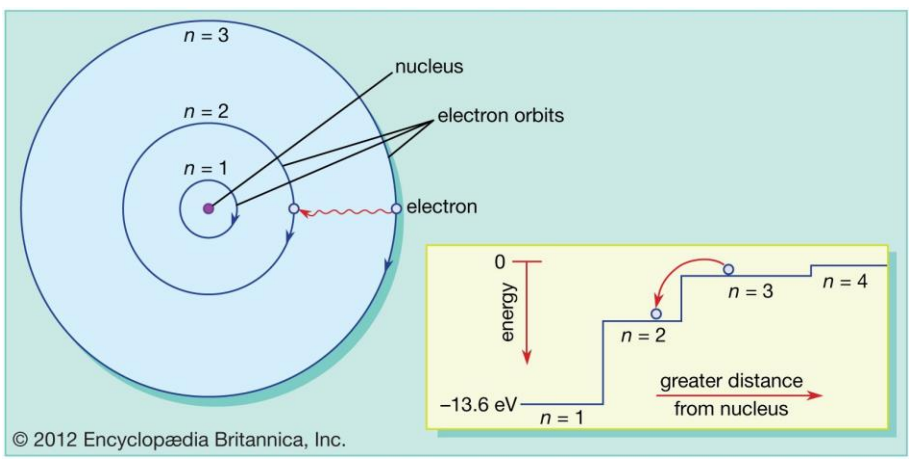




This model introduced the concept of the nucleus but could not explain atomic stability.

Bohr’s Model of Atom

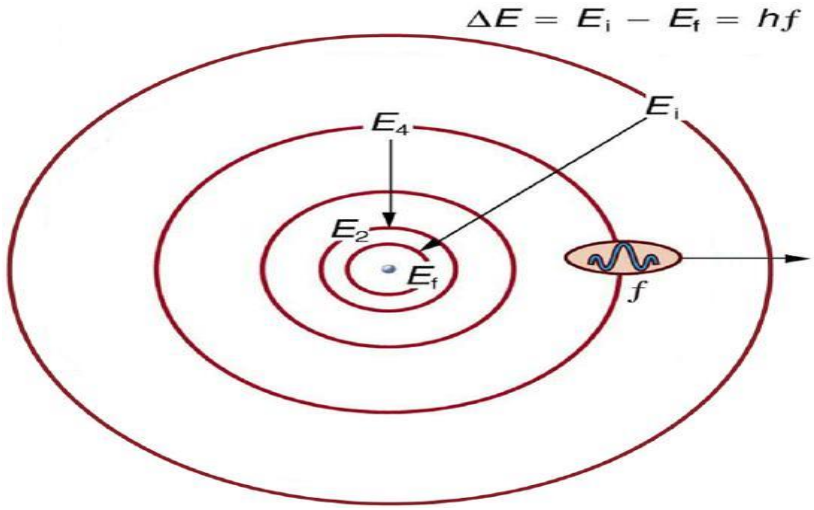
Bohr proposed that electrons revolve around the nucleus in fixed circular orbits with quantized energy levels. Electrons can move between these levels by absorbing or emitting energy.



This model successfully explained the hydrogen spectrum.

Bohr's Equations

The energy of an electron in the nth orbit and radius of orbit are given by:



$$E = E_f - E_i$$

$$E = \frac{1}{n_i^2} E_1 - \frac{1}{n_f^2} E_1 \;\; using \; E_n = \frac{1}{n^2} E_1$$

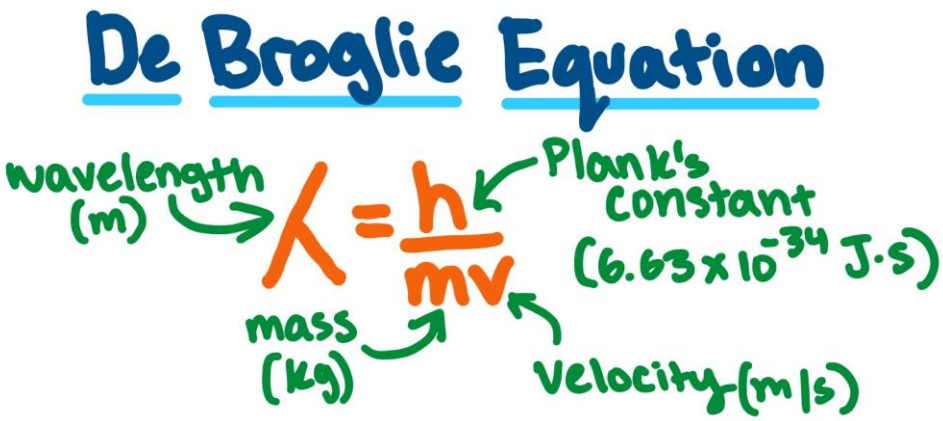
$$hf = E_1 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

These equations are important for numerical problems.

Dual Nature of Matter

According to de Broglie, matter exhibits both particle and wave nature. Every moving particle has an associated wavelength.



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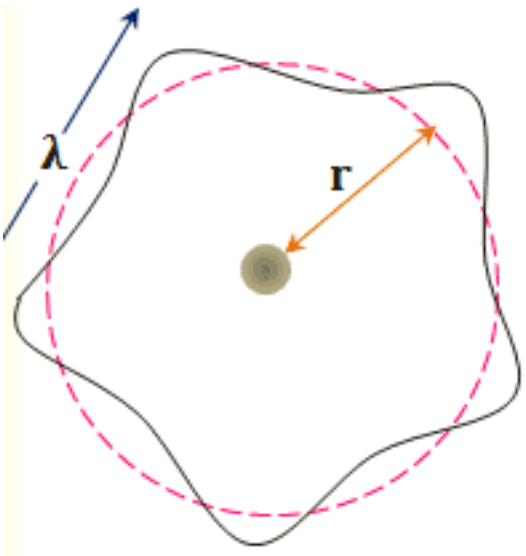
$$\lambda = \frac{h}{mv} = \frac{h}{p}$$



$$E = h\nu = h\frac{c}{\lambda}$$

$$E = mc^2$$

de Broglie relatio

$$\lambda = \frac{h}{mc} = \frac{h}{p}$$

This concept forms the basis of quantum mechanics.

Heisenberg's Uncertainty Principle

Heisenberg stated that it is impossible to simultaneously determine the exact position and momentum of a particle.

Heisenberg Uncertainty Relationships

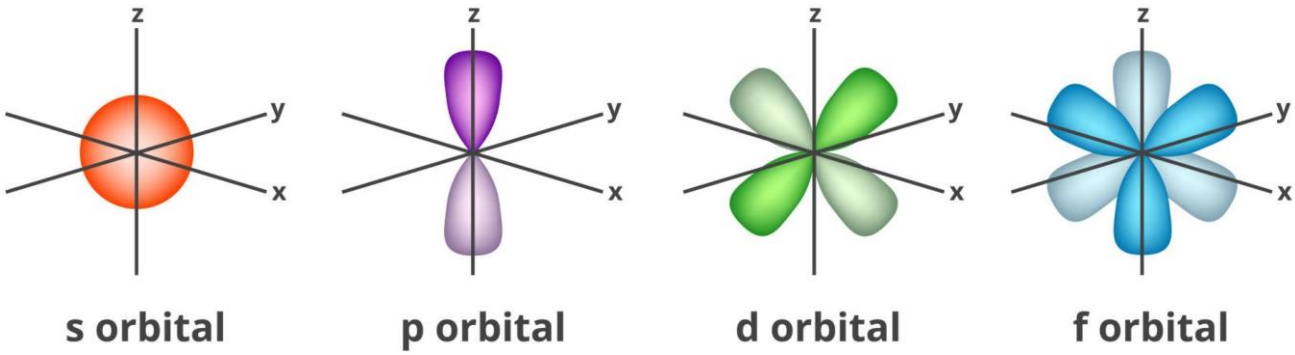
$$\Delta x \cdot \Delta p \sim \hbar$$

$$\Delta E \cdot \Delta t \sim \hbar$$

This principle limits precise measurement at microscopic level.

Quantum Mechanical Model

The quantum mechanical model describes electrons as wave functions rather than particles in fixed orbits. It introduces orbitals as regions of high probability of finding electrons.



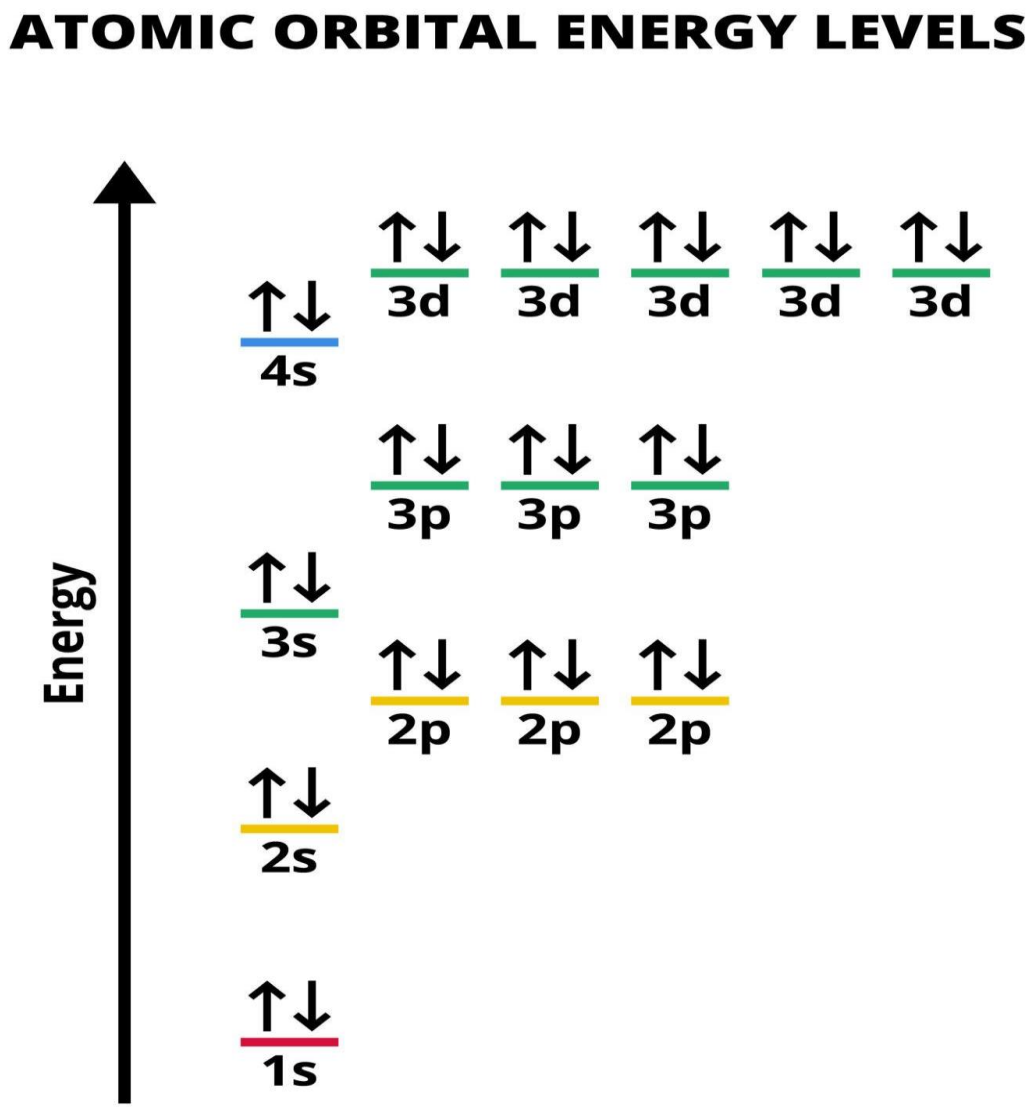
This model provides the most accurate description of atomic structure.

Quantum Numbers

Quantum numbers describe the position and energy of an electron in an atom. These include principal, azimuthal, magnetic, and spin quantum numbers.

Electronic Configuration

Electronic configuration refers to the arrangement of electrons in different orbitals. It follows principles such as Aufbau principle, Pauli exclusion principle, and Hund's rule.



These rules help determine the distribution of electrons.